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Data Article

Multi-domain vibration dataset with various bearing types under compound machine fault scenarios

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A R T I C L E I N F O

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Dataset link: Multi-domain vibration dataset with various bearing types under compound machine fault scenarios: subset 1 (deep groove ball bearing) (Original data)
Dataset link: Multi-domain vibration dataset with various bearing types under compound machine fault scenarios: subset 2 (cylindrical roller bearing) (Original data)
Dataset link: Multi-domain vibration dataset with various bearing types under compound machine fault scenarios: subset 3 (tapered roller bearing) (Original data)

Keywords:
Compound Fault
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A B S T R A C T

In modern complex mechanical systems, machine faults typically occur in multiple components simultaneously, and the domain of collected sensor data changes continuously due to variations in operating conditions. Deep learning-based fault diagnosis approaches have recently been enhanced to address these real-world industrial challenges. Comprehensive labeled data covering compound fault scenarios and multi-domain conditions are crucial for exploring these issues. However, existing multi-domain datasets focus on a limited range of operating conditions, such as motor rotating speeds and loads. This limits their applicability to real-world industrial scenarios. To bridge this gap, we present a novel multi-domain dataset that incorporates these basic conditions and extends to various bearing types and compound machine faults. The deep groove ball bearing, the cylindrical roller bearing, and the tapered roller bearing were utilized to provide data that reflect diverse mechanical interactions between the shaft and the bearing. Vibration data were collected using a USB digital accelerometer at two sampling rates and six rotating speeds, encompassing three single bearing faults, seven single rotating component faults, and 21 compound faults of the bearing and rotating component. Additionally, the dataset provides spectrograms of vibration data using short-time Fourier transform (STFT) for data-driven analysis with a 2-D input. This dataset encom-

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passes more complex compound fault and domain shift problems than those presented in conventional public vibration datasets, thereby aiding researchers in studying intelligent fault diagnosis methods based on deep learning.

Specifications Table

Subject	Mechanical engineering
Specific subject area	Compound fault diagnosis, domain adaptation, machine condition monitoring, vibration analysis.
Type of data	Raw, Processed (MAT file)
Data collection	Raw vibration data were gathered using a USB digital accelerometer (PCB Piezotronics 333D01) from the AST fault simulator at sampling rates of 8 kHz and 16 kHz. Looseness, unbalance, and misalignment were simulated for faults of rotating components. Three types of bearing were used to generate multi-domain datasets: a deep groove ball bearing (6204), cylindrical roller bearings (N204 and NJ204), and a tapered roller bearing (30204), each with the ball, inner-raceway, and outer-raceway faults. Additionally, data were collected at six different rotating speeds. The raw vibration signals were also converted to 2-D spectrograms using the short-time Fourier transform (STFT) by the MATLAB program.
Data source location	Institution: University of Seoul City/Town/Region: Seoul Country: South Korea
Data accessibility	Repository name: Multi-domain vibration dataset with various bearing types under compound machine fault scenarios: subset 1 (deep groove ball bearing) Data identification number: 10.17632/53vtnjy6c6 Direct URL to data: https://data.mendeley.com/datasets/53vtnjy6c6 Repository name: Multi-domain vibration dataset with various bearing types under compound machine fault scenarios: subset 2 (cylindrical roller bearing) Data identification number: 10.17632/7trwzz77xh Direct URL to data: https://data.mendeley.com/datasets/7trwzz77xh Repository name: Multi-domain vibration dataset with various bearing types under compound machine fault scenarios: subset 3 (tapered roller bearing) Data identification number: 10.17632/2cygy6y4rk Direct URL to data: https://data.mendeley.com/datasets/2cygy6y4rk

1. Value of the Data

- Unlike conventional bearing fault datasets, which are limited to single-bearing faults [1,2], this dataset facilitates the analysis of two complex machine fault scenarios: compound faults and domain shift.
- Our dataset includes 21 different compound faults of rotating machinery. Researchers can leverage the dataset to tackle practical engineering scenarios involving the simultaneous failure of multiple rotating machinery components often encountered in real industrial settings.
- By providing vibration signals from three different types of bearings, our dataset captures significant domain shift scenarios in which the kinematic properties of rotating machinery change. Researchers and engineers can utilize this dataset to investigate state-of-the-art methods aiming at domain adaptation or continual learning within these challenging, real-world domain shift environments.
- Our dataset also provides the time-frequency domain data preprocessed using short-time Fourier transform (STFT) to assist research on data-driven fault diagnosis methods using 2-D inputs, such as 2-D convolutional neural networks.

2. Background

Owing to their superior fault identification performance compared to conventional model-driven or machine learning-based approaches, deep learning-based data-driven fault diagnosis approaches have gained significant attention in recent years. Many researchers have adopted these methods to address complex mechanical systems, in which failures can often occur in multiple components simultaneously, referred to as compound faults [3]. Additionally, the operating conditions of machines can change dynamically, leading to shifts in the domain of collected sensor data [4]. Previous machine fault datasets primarily addressed domain shift problems by providing data under various rotating speed conditions [5–7] but lacked diversity in the characteristics of the mechanical component in the test rig. Although some studies [8,9] collected vibration signals from different bearing models, their data only included bearings with the same contact types as deep groove ball bearings. To overcome these limitations, our dataset offers vibration data collected from a machine with compound faults and three different bearing types, allowing researchers to explore a broader range of domain shift problems using compound fault data.

3. Data Description

Data were collected under two sampling rates, three bearing types, and six rotating speeds, including bearing faults, rotating component faults, and compound faults of bearing and rotating components. Bearing faults encompass ball (B), inner-raceway (IR), and outer-raceway (OR) faults. For the rotating component, looseness (L), unbalance (U), and misalignment (M) faults were simulated. U and M faults were further categorized into three severity levels, designated by a postfix number following the fault name, such as U1, U2, and U3, where a higher number indicates a more significant severity level of the fault. A compound fault is a combination of bearing and rotating component faults. Consequently, the total number of machine conditions is 32, comprising one healthy state, three single bearing faults, seven single rotating component faults, and 21 compound faults. The healthy condition, denoted by H, represents the condition where no faults are present in either the bearing or rotating component.

The dataset is stored in three subset repositories according to the bearing type, as outlined in the specifications table section. Each subset is hierarchically structured according to the sampling rate and the rotating speed. Fig. 1 illustrates the hierarchical structure for storing data files, where each directory corresponds to a specific sampling rate and rotating speed. In each directory, data samples are stored in binary MATLAB (MAT) files. The naming convention of these data files follows the format: *{rotating component condition}_{bearing condition}_{sampling rate}_{bearing model}_{rotating speed}.mat*. Table 1 lists the naming convention of each property of the data file name. For instance, the data file name of *H_IR_8_6204_1000.mat* indicates that data samples were collected at an 8 kHz sampling rate from a machine with a healthy rotating

Table 1
Naming convention of each property in the data file name.

Property	Name	Unit
rotating component condition	H, L, U1, U2, U3, M1, M2, or M3	-
bearing condition	H, B, IR, or OR	-
sampling rate	8 or 16	kHz
bearing model	6204[1], N204[2], NJ204[2], or 30204[3]	-
rotating speed	600, 800, 1000, 1200, 1400, or 1600	RPM

[1] Deep groove ball bearing
[2] Cylindrical roller bearings
[3] Tapered roller bearing.

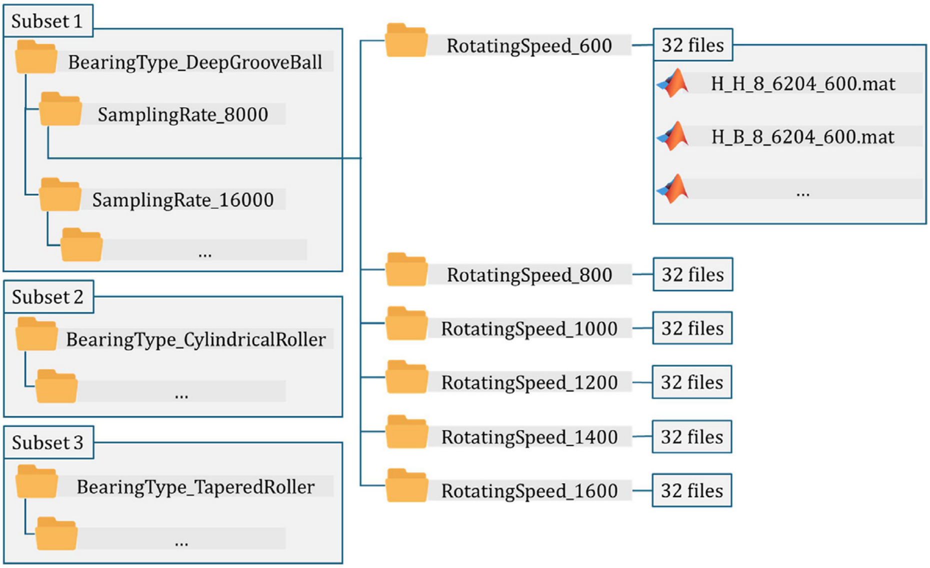

Fig. 1. Directory hierarchy of dataset.

component, bearing model 6204 with the IR fault, and a rotating speed of 1,000 revolutions per minute (RPM).

Each MAT file has four data fields: *Data, Spectrogram, STFTFreq*, and *STFTTime*. The *Data* field contains raw vibration data samples collected from the uniaxial time domain. The unit of the data value is gravitational acceleration g (9.80665 m/s^2), and the number of data points is 1,280,000. The *Spectrogram* field stores spectrograms as a 3-D array with dimensions $N_{spectrogram} \times N_{freq} \times N_{time}$, where $N_{spectrogram}$, N_{freq}, and N_{time} denote the number of spectrograms, the size of the frequency domain axis, and the size of the time domain axis, respectively. The unit of the spectrogram value is the dB scale amplitude, and the reference acceleration used for dB scale conversion is 1 g, matching the unit of the raw vibration data. The spectrogram size of our dataset is 78×128×128. The *STFTFreq* and *STFTTime* fields hold the frequency and time instant vectors of the STFT output, respectively. The unit of the frequency vector is Hz, and the unit of the time instant vector is seconds. Fig. 2(a) and (b) illustrate the raw vibration signal and its corresponding spectrogram with a 16 kHz sampling rate, a 1,000 RPM rotating speed, and the compound fault of IR and M2 for three bearing types, respectively.

4. Experimental Design, Materials and Methods

The AST fault simulator, which consists of a brushless DC (BLDC) motor, a shaft, and two rotor disks, was used as a data collection testbed. The components of the simulator are supported by two bearings and their housings. The BLDC motor has a 40 W power output, a 60 Hz frequency, and a maximum rotating speed of 1,700 RPM. Data were collected at rotating speeds of 600, 800, 1,000, 1,200, 1,400, and 1,600 RPMs. The rotating speed of the BLDC motor is measured by a Hall sensor and can be adjusted using a dial button.

For data collection, a PCB Piezotronics 333D01 accelerometer was utilized. It was mounted with a magnetic stud on the top side of the bearing housing at the shaft end and was connected to a laptop computer via a USB interface. The sensitivity and measurement range of the accelerometer are 4.00 % FSV/g and ± 20 g, respectively. The accelerometer recorded data at 8 kHz and 16 kHz sampling rates. To obtain an equal number of data points for both sampling

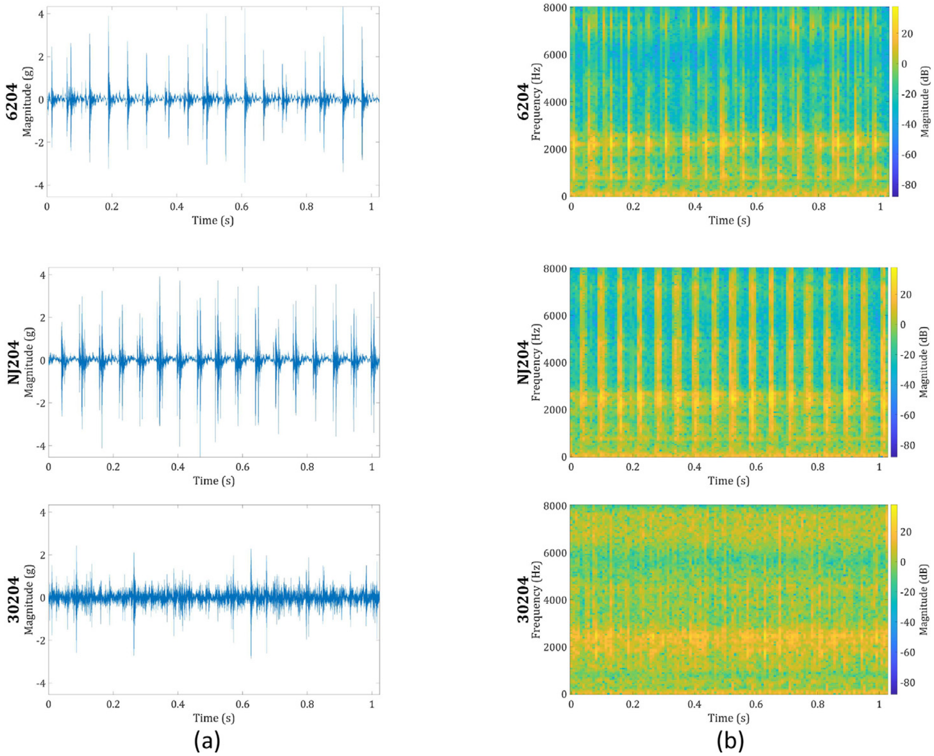

Fig. 2. Visualization of vibration data collected from IR and M2 faults at 16 kHz sampling rate and 1,000 RPM rotating speed for three bearing types. (a) time domain data. (b) corresponding spectrogram.

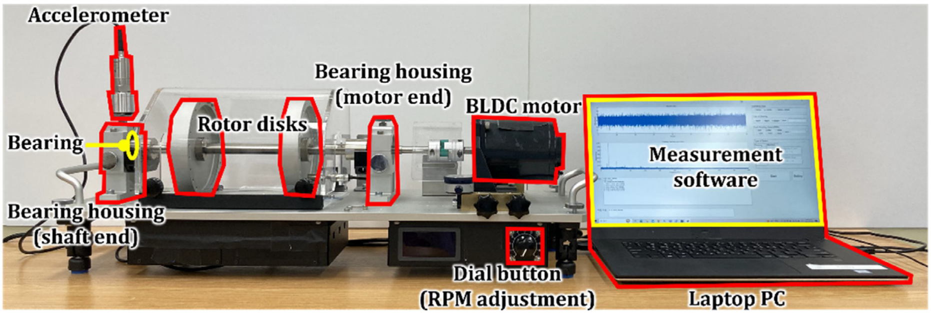

Fig. 3. Front view of testbed used for vibration data collection.

rates, data samples were collected for 160 s at 8 kHz and 80 s at 16 kHz. Fig. 3 shows the testbed setup for data collection.

Three different types of rotating component faults were artificially simulated. As shown in Fig. 4(a), an L fault was generated by loosening the screw of the bearing housing located at the motor end for a half-turn. For M faults of severity levels 1, 2, and 3, the central axis of the BLDC motor was shifted by 0.6 mm, 0.8 mm, and 1.0 mm, respectively, using the adjustment thread shown in Fig. 4(b). A U fault was simulated by attaching an additional screw to the rotor disk; for severity levels 1, 2, and 3, 3 g, 4 g, and 5 g of additional mass were attached, respectively. Fig. 4(c) shows the rotor disk with the additional mass.

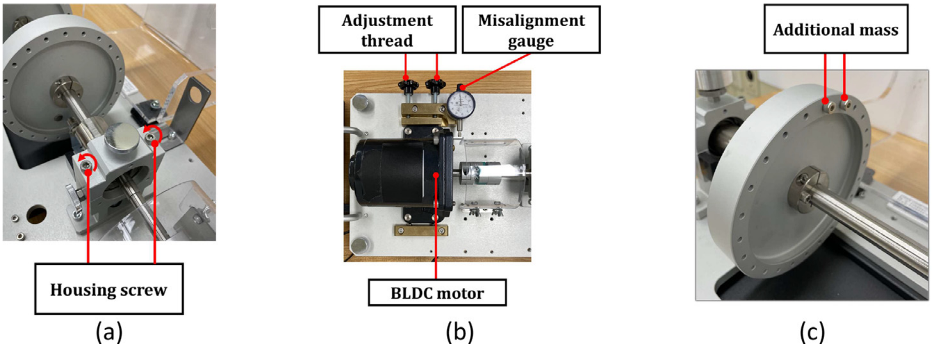

Fig. 4. Simulation of rotating component faults. (a) Bearing housing unscrewing for the L fault. (b) adjustment thread for the M fault. (c) screws attached to the rotor disk for the U fault.

Table 2
Bearing model specifications.

Bearing model	Bearing type	Pitch diameter (mm)	Ball diameter (mm)	Contact angle (°)	Number of rolling elements
6204	Deep groove ball bearing	34.57	7.94	0	8
N204 NJ204	Cylindrical roller bearing	34	7.50	0	11
30204	Tapered roller bearing	35.8	6.20	12.6	15

The following three types of bearings were utilized to provide multi-domain data: the deep groove ball bearing, the cylindrical roller bearing, and the tapered roller bearing. The specifications of the bearing models used in this dataset are summarized in Table 2. The N204 and NJ204 bearing models share the same contact type and have nearly identical specifications. However, a primary difference between them is that the N204 bearing can separate its outer raceway, whereas the NJ204 bearing can separate its inner raceway; therefore, the N204 and NJ204 bearings were used for simulating the B and OR faults and the IR fault, respectively. Defective bearings were manufactured using the grinding method and installed in the bearing housing at the shaft end of the testbed. Fig. 5 shows the defective bearings exhibiting B, IR, and OR faults.

Data samples were collected using MATLAB-based in-house software. The software supports data visualization in the time and frequency domains and metadata annotation, such as fault types and rotating speeds. Fig. 6 illustrates a snapshot of our data measurement software. The data sample collection procedure was performed as follows:

1. Install the accelerometer at the top of the shaft end housing.
2. Install the bearing at the shaft end housing.
3. Set up the rotating component fault.
4. Activate the fault simulator.
5. Adjust the rotating speed using the dial button.
6. Collect the data samples using the measurement software.
7. Save the data as a MAT file.
8. Deactivate the fault simulator.
9. Repeat 2–8 for all scenarios.

To ensure the integrity of all collected data, a two-step validation process was conducted. First, sensor errors, such as out-of-range measurements and invalid segments, as well as noise interference were addressed. All sensor readings were checked to ensure that they were within the measurement range of the accelerometer, i.e., ±20 *g*, and contained valid floating-point values. While none of the data values exceeded the measurement range, occasional errors produced

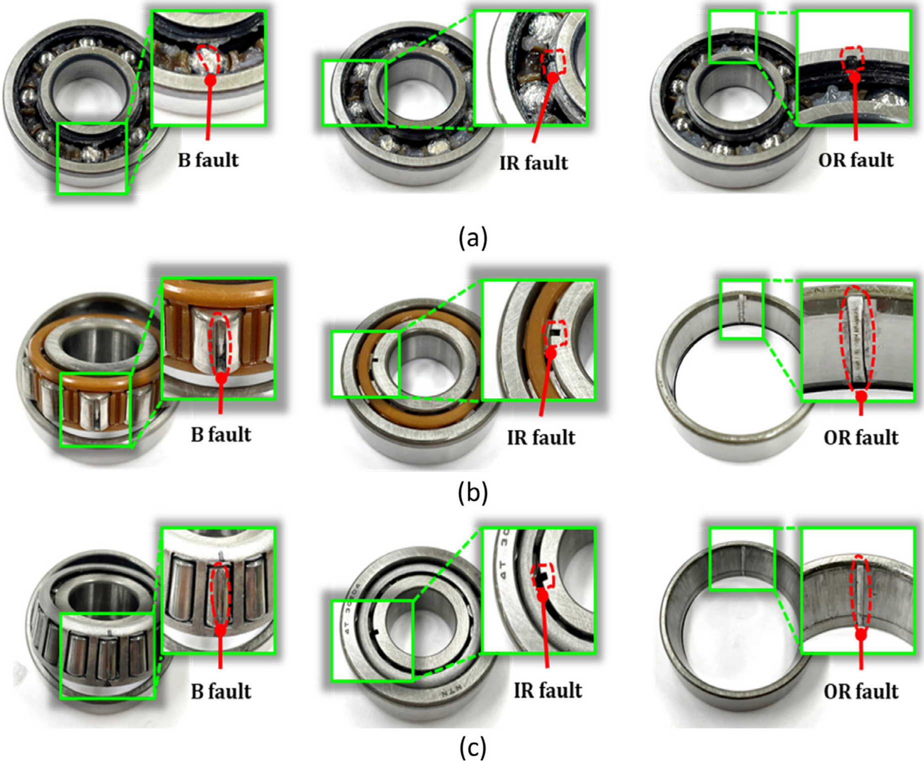

Fig. 5. Defective bearings. (a) 6204 bearing (B, IR, and OR faults). (b) N204 (B and OR faults) and NJ204 (IR fault) bearings. (c) 30204 bearing (B, IR, and OR faults).

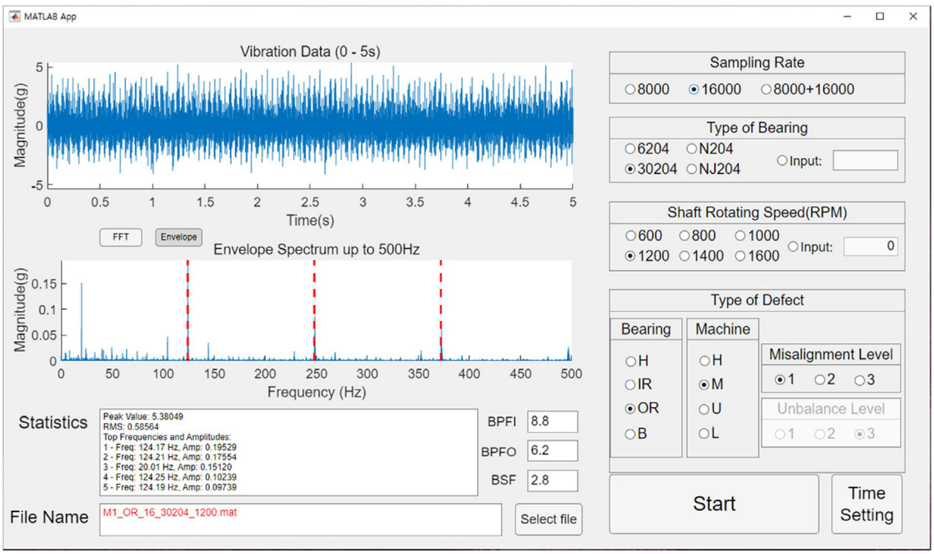

Fig. 6. Snapshot of data measurement software.

invalid segments, i.e., NaN values. In such cases, the data were recollected. The presence of noise in the data was assessed by examining the visualized raw and frequency domain signals. Signals with abnormally high amplitudes in frequency ranges outside the machine's rotational frequency, bearing fault frequencies, or their harmonics were regarded as noisy. Although the data were collected in a controlled laboratory environment, noise interference occasionally occurred. Whenever this issue was identified, the data were recollected.

Second, to verify that the simulated faults accurately represent real-world fault conditions, bearing fault frequency and signal amplitude were evaluated. For data with bearing faults, the frequency domain of the envelope signal was examined to identify the presence of peaks at the corresponding fault frequencies, such as ballpass frequency inner race (BPFI), ballpass frequency outer race (BPFO), and ball spin frequency (BSF). In the case of machine faults, the signal amplitude of the raw vibration signal, such as peak or root-mean-square (RMS), was analyzed to confirm that the vibration level of machine fault data was higher than that of healthy data. The validation results demonstrated that the faulty data in our dataset accurately represented the characteristics of actual faults.

After data validation, spectrograms were generated from the original data using the STFT method with the Kaiser window function. The Kaiser window function was selected for its flexibility in adjusting the trade-off between frequency resolution and spectrum leakage caused by signal noise by tuning the parameter β. A value of $\beta = 0.5$ was used in this dataset, which is relatively low, to enhance frequency resolution. Although a lower β value allows for more spectrum leakage, this setting is appropriate for our controlled data collection environment with minimal external noise interference. A sample size of 16,384 was chosen to ensure that the spectrograms captured at least ten rotations, providing sufficient observations for fault detection across all sampling rates and rotating speeds. The window size and overlap size were set to 192 and 65, respectively, considering the balance between the time resolution and the spectrogram size. For example, the time resolution of the spectrogram with an 8 kHz sampling rate is 15.875 ms, which can capture frequency changes at approximately twice the maximum rotating speed of 1,600 RPM, and the spectrogram size is 128×128. In addition, an FFT size of 256 was chosen, which is the nearest power of 2 greater than the window size.

Limitations

The faults of the bearings and rotating components were generated artificially in a laboratory environment. Although the dataset was validated to reflect the characteristics of actual bearing and machine faults, it may not fully encompass real-world industrial conditions, such as external vibration or electromagnetic noise from other machines.

Ethics Statement

This work does not involve human subjects, animal experiments, or any data collected from social media platforms.

Data Availability

Multi-domain vibration dataset with various bearing types under compound machine fault scenarios: subset 1 (deep groove ball bearing) (Original data)

Multi-domain vibration dataset with various bearing types under compound machine fault scenarios: subset 2 (cylindrical roller bearing) (Original data)

Multi-domain vibration dataset with various bearing types under compound machine fault scenarios: subset 3 (tapered roller bearing) (Original data)

CRediT Author Statement

Seongjae Lee: Conceptualization, Methodology, Validation, Writing – original draft, Writing – review & editing; **Taewan Kim:** Methodology, Software, Visualization, Data curation, Writing – original draft, Writing – review & editing; **Taehyoun Kim:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing, Supervision, Funding acquisition.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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